

$p \rightarrow q$ is a tautology.

Worked Examples 1.1

Example 1.1.1

If p and q are primitive statements for which the implication $p \rightarrow q$ is false. Determine the truth value for the following;

- a) $p \wedge q$ b) $\neg p \vee q$ c) $\neg q \rightarrow \neg p$

Solution:

Since $p \rightarrow q$ is false, the truth value of p is true and the truth value of q is false.

a) Since the truth value of p is true and q is false, the truth value of $p \wedge q$ is false.

b) Since the truth value of p is true, the truth value of $\neg p$ is false. Therefore, the truth value of $\neg p \vee q$ is false.

c) The truth value of $\neg q$ is true and $\neg p$ is false. Therefore, the truth value of $\neg q \rightarrow \neg p$ is false.

Example 1.1.2

Construct a truth table for the compound statement $\neg(p \vee \neg q) \rightarrow \neg p$ where p and q are primitive statements. Check whether it is a tautology.

Solution:

p	q	$\neg q$	$p \vee \neg q$	$\neg(p \vee \neg q)$	$\neg p$	$\neg(p \vee \neg q) \rightarrow \neg p$
1	1	0	1	0	0	1
1	0	1	1	0	0	1
0	1	0	0	1	1	1
0	0	1	1	0	1	1

Table:1.6

Since the truth value of the expression $\neg(p \vee \neg q) \rightarrow \neg p$ is true for all possible assignments of p and q , the proposition is a tautology.

Example 1.1.3

Construct a truth table and show that implication $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology.

Solution:

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$[p \wedge (p \rightarrow q)] \rightarrow q$
1	1	1	1	1
1	0	0	0	1
0	1	1	0	1
0	0	1	0	1

Table:1.7

From the truth table it can be seen that the expression $[p \wedge (p \rightarrow q)] \rightarrow q$ is true that for all assigned values of p and q . Therefore, it is a tautology.

Example 1.1.4

Use truth table to verify that the expression $\neg(p \rightarrow q) \Rightarrow \neg q$ is a logical implication.

Solution:

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$	$\neg q$	$\neg(p \rightarrow q) \rightarrow \neg q$
1	1	1	0	0	1
1	0	0	1	1	1
0	1	1	0	0	1
0	0	1	0	1	1

Table:1.8

Since the expression $\neg(p \rightarrow q) \rightarrow \neg q$ is a tautology, the implication is a logical implication.

Example 1.1.5

Check whether the implication $p \rightarrow (q \rightarrow r)$ is a tautology by constructing a truth table.

Solution:

p	q	r	$q \rightarrow r$	$p \rightarrow (q \rightarrow r)$
1	1	1	1	1
1	0	0	1	1
0	1	0	0	1
0	0	1	1	1
1	1	0	0	0
1	0	1	1	1
0	1	1	1	1
0	0	0	1	1

Table:1.9

Since the truth value of $p \rightarrow (q \rightarrow r)$ is not true for some assignment of p and q , it is not a tautology.

Example 1.1.6

Construct a truth table for the compound statement $q \leftrightarrow (\neg p \vee \neg q)$ where p and q are primitive statements. Check whether it is a tautology.

Solution:

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$	$q \leftrightarrow (\neg p \vee \neg q)$
1	1	0	0	0	0
1	0	0	1	1	0
0	1	1	0	1	1
0	0	1	1	1	0

Table:1.10

Since the truth value of $q \leftrightarrow (\neg p \vee \neg q)$ is not true for some assignments of p and q , it is not a tautology.

Example 1.1.7

Show that the statement $p \wedge (\neg p \wedge q)$ is a contradiction where p and q are primitive statements.

Solution:

p	q	$\neg p$	$\neg p \wedge q$	$p \wedge (\neg p \wedge q)$
1	1	0	0	0
1	0	0	0	0
0	1	1	1	0
0	0	1	0	0

Table:1.11

From the truth table it can be seen that for all possible value of p and q the expression $p \wedge (\neg p \wedge q)$ is false. Therefore, it is a contradiction.

Example 1.1.8

Show that the following expression $q \vee (p \vee \neg q)$ is a tautology by constructing a truth table.

Solution:

p	q	$\neg q$	$p \vee \neg q$	$q \vee (p \vee \neg q)$
1	1	0	1	1
1	0	1	1	1
0	1	0	0	1
0	0	1	1	1

Table:1.12

The truth value of the expression $q \vee (p \vee \neg q)$ is true for all possible assignments of p and q $\therefore$ it is a tautology.

Example 1.1.9

Construct a truth table and check whether implication $((p \rightarrow q) \wedge (q \rightarrow r)) \Rightarrow (p \rightarrow r)$ is a logical implication.

Solution:

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
1	1	1	1	1	1	1	1
1	0	0	0	1	0	0	1
0	1	0	1	0	0	1	1
0	0	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	0	1	1
0	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1

Table:1.13

Since $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$ is a tautology, the implication is a logical implication.

Example 1.1.10

Construct a truth table for the compound statement $(\neg(p \wedge q) \vee r) \rightarrow \neg p$, where p and q are primitive statements.

Solution:

p	q	r	$p \wedge q$	$\neg(p \wedge q)$	$\neg(p \wedge q) \vee r$	$\neg p$	$(\neg(p \wedge q) \vee r) \rightarrow \neg p$
1	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
0	1	0	0	1	1	1	1
0	0	1	0	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	1	0	0
0	1	1	0	1	1	1	1
0	0	0	0	1	1	1	1

Table:1.14

The truth value of the expression $(\neg(p \wedge q) \vee r) \rightarrow \neg p$ is not true for some assignments of p and q $\therefore$ it is not a tautology.

Example 1.1.11

Show that the following expression $q \vee (p \vee \neg q)$ is a tautology by constructing a truth table.

Solution:

p	q	$\neg q$	$p \vee \neg q$	$q \vee (p \vee \neg q)$
1	1	0	1	1
1	0	1	1	1
0	1	0	0	1
0	0	1	1	1

Table:1.15

The truth value of the expression $q \vee (p \vee \neg q)$ is true for all possible assignments of p and q $\therefore$ it is a tautology.

Example 1.1.12

Prove that the propositions $(p \rightarrow q) \Leftrightarrow \neg p \vee q$ are logically equivalent.

Solution:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
1	1	1	0	1
1	0	0	0	0
0	1	1	1	1
0	0	1	1	1

Table:1.16

The truth values of $p \rightarrow q$ and $\neg p \vee q$ are same. Therefore, the propositions are logically equivalent.

Example 1.1.13

Prove that the propositions $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$ are logically equivalent.

Solution:

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q$	$(p \rightarrow q) \wedge (q \rightarrow p)$
1	1	1	1	1	1
1	0	0	1	0	0
0	1	1	0	0	0
0	0	1	1	1	1

Table:1.17

The truth values of $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are same. Therefore, the propositions are logically equivalent.

Example 1.1.14

Check whether the propositions $p \wedge (\neg q \vee r)$ and $p \vee (q \wedge \neg r)$ are logically equivalent.

Solution:

p	q	r	$\neg q$	$\neg q \vee r$	$p \wedge (\neg q \vee r)$	$\neg r$	$q \wedge \neg r$	$(p \vee (q \wedge \neg r))$
1	1	1	0	1	1	0	0	1
1	0	0	1	1	1	1	0	1
0	1	0	0	0	0	1	1	1
0	0	1	1	1	0	0	0	0
1	1	0	0	0	0	1	1	1
1	0	1	1	1	1	0	0	1
0	1	1	0	1	0	0	0	0
0	0	0	1	1	0	1	0	0

Table:1.18

The truth values of $p \wedge (\neg q \vee r)$ and $p \vee (q \wedge \neg r)$ are not same. Therefore, the propositions are logically not equivalent.

Example 1.1.15

Check whether the propositions $p \vee (q \wedge r)$ and $(p \vee q) \wedge r$ are logically equivalent.

Solution:

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$(p \vee q) \wedge r$
1	1	1	1	1	1	1
1	0	0	0	1	1	0
0	1	0	0	0	1	0
0	0	1	0	0	0	0
1	1	0	0	1	1	0
1	0	1	0	1	1	1
0	1	1	1	1	1	1
0	0	0	0	0	0	0

Table:1.19

The truth values of $p \vee (q \wedge r)$ and $(p \vee q) \wedge r$ are not same. Therefore, the propositions are logically not equivalent.

Example 1.1.16

Determine all truth value assignments, if any, for the primitive statements p, q, r, s, t that make the statement $[(p \wedge q) \wedge r] \rightarrow (s \vee t)$ false.

Solution:

$[(p \wedge q) \wedge r] \rightarrow (s \vee t)$ is false when $(s \vee t)$ is false and $(p \wedge q) \wedge r$ is true. $(p \wedge q) \wedge r$ is true when p, q, r are true. $s \vee t$ is false when s and t are false.

Example 1.1.17

Determine the truth value of the implication, If $3 + 3 = 6$ then $3 + 4 = 9$.

Solution:

Let $p: 3 + 3 = 6$, $q: 3 + 4 = 9$. The truth value of p is true and the truth value of q is false. Then the truth value of $p \rightarrow q$ is false.

Example 1.1.18

If statement q has a truth value 1, determine all truth value assignment for the primitive statements p, r and s for which the truth value of the statement

$$[q \rightarrow ((\neg p \vee r) \wedge \neg s)] \wedge [\neg s \rightarrow (\neg r \wedge q)] \text{ is } 1.$$

Solution:

Since the truth value of the given statement is true, the truth value of

$[q \rightarrow ((\neg p \vee r) \wedge \neg s)]$ is 1 and the truth value of $[\neg s \rightarrow (\neg r \wedge q)]$ is 1. Given q has a truth value 1, $[q \rightarrow ((\neg p \vee r) \wedge \neg s)]$ is true when the truth value of $((\neg p \vee r) \wedge \neg s)$ is 1. Which is possible when the truth value of $\neg p \vee r$ and the truth value of $\neg s$ is 1. So, the truth value of s is 0. Then the truth value of $\neg s \rightarrow (\neg r \wedge q)$ is 1 when the truth value of $\neg r \wedge q$ is 1. Since q has a truth value 1, the truth value of $\neg r$ is 1. Therefore, the truth value of r is 0. Since the truth value of $\neg p \vee r$ is true, the truth value of $\neg p$ is true. Therefore the truth value of p is 0.

Exercises.1.1

1. Construct a truth table for each of the following compound statements, where p, q, r denote the primitive statements. Check whether it is a tautology.

- a) $(p \wedge q) \rightarrow p$ b) $\neg(p \rightarrow q) \rightarrow \neg p$
 c) $\neg p \vee q \rightarrow \neg q$ d) $((p \rightarrow q) \wedge [(q \wedge \neg r)] \rightarrow (p \vee r))$ e) $(p \rightarrow q) \rightarrow (q \rightarrow p)$
 f) $q \wedge (\neg r \rightarrow p)$

2. Check whether the propositions are logically equivalent or not,

- a) $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$
 b) $p \rightarrow (q \wedge r)$ and $(p \rightarrow q) \wedge (p \rightarrow r)$ c) $[p \rightarrow (q \vee r)] \Leftrightarrow [\neg r \rightarrow (p \rightarrow q)]$

3. For primitive statements p and q verify that $p \rightarrow [q \rightarrow (p \wedge q)]$ is a tautology.

4. Determine the truth value of the implication, 'If $3 + 4 = 12$ then $3 + 2 = 6$.'

5. Assuming the truth value of p and r are false and that of q and s are true, find the truth value of the proposition $((p \wedge \neg q) \rightarrow (q \wedge r)) \rightarrow (s \vee \neg q)$.

Answers: 1.1

1. (a) Tautology b) Not a tautology c) Not a tautology d) Not a tautology e) Not a Tautology f) Not a tautology
 2. a) not logically equivalent b) logically equivalent c) logically equivalent
 4. True. 5. True

Example 1.2.1

Prove that $[(p \vee q) \wedge ((p \vee \neg q))] \vee q \Leftrightarrow p \vee q$

Solution:

$[(p \vee q) \wedge ((p \vee \neg q))] \vee q$	
$\Leftrightarrow [(p \vee (q \wedge \neg q))] \vee q$	Distributive Law
$\Leftrightarrow (p \vee F_0) \vee q$	Inverse Law
$\Leftrightarrow (p \vee q)$	Identity Law

Table:1.21

Example 1.2.2

Prove that the statements are logically equivalent without using truth table

$$(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$$

Solution:

$(p \vee q) \wedge \neg(\neg p \wedge q)$	
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q)$	Demorgan's Law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Law of double negation
$\Leftrightarrow p \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow p \vee F_0$	Inverse Law
$\Leftrightarrow p$	Identity Law

Table:1.22

Example 1.2.3

Prove that $(p \rightarrow q) \wedge (r \rightarrow q) \Leftrightarrow (p \vee r) \rightarrow q$

Solution:

$(p \rightarrow q) \wedge (r \rightarrow q)$	
$\Leftrightarrow (\neg p \vee q) \wedge (\neg r \vee q)$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg r) \vee q$	Distributive Law
$\Leftrightarrow \neg(p \vee r) \vee q$	Demorgan's Law
$\Leftrightarrow p \vee r \rightarrow q$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$

Example 1.2.4

Prove that $(p \vee q) \wedge (\neg(\neg p \wedge q)) \Leftrightarrow p$

Solution:

$(p \vee q) \wedge (\neg(\neg p \wedge q)) \Leftrightarrow p$	
$\Leftrightarrow (p \vee q) \wedge (\neg\neg p \vee \neg q)$	Demorgan's Law
$\Leftrightarrow (p \vee q) \wedge (p \vee \neg q)$	Law of double negation
$\Leftrightarrow (p \vee (q \wedge \neg q))$	Distributive Law
$\Leftrightarrow (p \vee F_0)$	Inverse Law
$\Leftrightarrow p$	Identity Law

Example 1.2.5

Prove that $(p \rightarrow q) \wedge \neg q \Leftrightarrow \neg(p \vee q)$

Solution:

$(p \rightarrow q) \wedge \neg q$	
$\Leftrightarrow (\neg p \vee q) \wedge \neg q$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg q) \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow (\neg p \wedge \neg q) \vee F_0$	Inverse Law
$\Leftrightarrow (\neg p \wedge \neg q)$	Identity Law
$\Leftrightarrow \neg(p \vee q)$	Demorgan's Law

Example 1.2.6

Prove that $(p \rightarrow q) \wedge \neg q \Leftrightarrow \neg(p \vee q)$

Solution:

$(p \rightarrow q) \wedge \neg q$	
$\Leftrightarrow (\neg p \vee q) \wedge \neg q$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \wedge \neg q) \vee (q \wedge \neg q)$	Distributive Law
$\Leftrightarrow (\neg p \wedge \neg q) \vee F_0$	Inverse Law
$\Leftrightarrow (\neg p \wedge \neg q)$	Identity Law
$\Leftrightarrow \neg(p \vee q)$	Demorgan's Law

Example 1.2.7

Without using truth table, show that $p \rightarrow (q \rightarrow r) \Leftrightarrow p \rightarrow (\neg q \vee r) \Leftrightarrow (p \wedge q) \rightarrow r$

Solution:

$p \rightarrow (q \rightarrow r)$	
$\Leftrightarrow p \rightarrow (\neg q \vee r)$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow \neg p \vee (\neg q \vee r)$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$
$\Leftrightarrow (\neg p \vee \neg q) \vee r$	Associative Law
$\Leftrightarrow \neg(p \wedge q) \vee r$	Demorgan's Law
$\Leftrightarrow (p \wedge q) \rightarrow r$	Since $(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$

Table:1.27

Example 1.2.8

Prove that $\neg[\neg((p \vee q) \wedge r) \vee \neg q] \Leftrightarrow q \wedge r$

Solution:

$\neg[\neg((p \vee q) \wedge r) \vee \neg q]$	
$\Leftrightarrow [\neg\neg((p \vee q) \wedge r) \wedge \neg\neg q]$	Demorgan's Law
$\Leftrightarrow [(p \vee q) \wedge r] \wedge q$	Law of double negation
$\Leftrightarrow (p \vee q) \wedge (r \wedge q)$	Associative Law
$\Leftrightarrow (p \vee q) \wedge (q \wedge r)$	Commutative Law
$\Leftrightarrow ((p \vee q) \wedge q) \wedge r$	Associative Law
$\Leftrightarrow (q \wedge (p \vee q)) \wedge r$	Commutative Law
$\Leftrightarrow q \wedge r$	Absorption Law

Table:1.28

Example 1.2.9

Prove that $(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \Leftrightarrow (\neg p \wedge q)$

Solution:

$(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q))$	
$\Leftrightarrow (p \vee q) \wedge ((\neg p \wedge \neg p) \wedge q)$	Associative Law
$\Leftrightarrow (p \vee q) \wedge (\neg p \wedge q)$	Idempotent Law
$\Leftrightarrow [p \wedge (\neg p \wedge q)] \vee [q \wedge (\neg p \wedge q)]$	Distributive Law
$\Leftrightarrow [(p \wedge \neg p) \wedge q] \vee [(q \wedge \neg p) \wedge q]$	Associative Law
$\Leftrightarrow ((p \wedge \neg p) \vee (q \wedge \neg p)) \wedge q$	Distributive Law
$\Leftrightarrow (F_0 \vee (q \wedge \neg p)) \wedge q$	Inverse Law
$\Leftrightarrow (q \wedge \neg p) \wedge q$	Identity Law
$\Leftrightarrow (\neg p \wedge q) \wedge q$	Commutative Law
$\Leftrightarrow \neg p \wedge (q \wedge q)$	Associative Law
$\Leftrightarrow (\neg p \wedge q)$	Idempotent Law

Table:1.29

Example 1.2.10

Prove that $(\neg p \wedge (\neg q \wedge r) \vee (q \wedge r) \vee (p \wedge r)) \Leftrightarrow r$

Solution:

$(\neg p \wedge (\neg q \wedge r) \vee (q \wedge r) \vee (p \wedge r))$	
$\Leftrightarrow [(\neg p \wedge \neg q) \wedge r] \vee (q \wedge r) \vee (p \wedge r)$	Associative Law
$\Leftrightarrow [\neg(p \vee q) \wedge r] \vee (q \wedge r) \vee (p \wedge r)$	Demorgan's Law
$\Leftrightarrow [\neg(p \vee q) \wedge r] \vee [(q \vee p) \wedge r]$	Distributive Law
$\Leftrightarrow [\neg(p \vee q) \vee (q \vee p)] \wedge r$	Distributive Law
$\Leftrightarrow [\neg(p \vee q) \vee (p \vee q)] \wedge r$	Commutative Law
$\Leftrightarrow T_0 \wedge r$	Inverse Law
$\Leftrightarrow r$	Identity Law

Table:1.30

Example 1.2.11

Write the Dual of a) $q \rightarrow r$ b) $p \rightarrow q \vee r$ c) $p \leftrightarrow q$ where p, q and r are primitive statements.

Solution:

a) $q \rightarrow r \Leftrightarrow \neg q \vee r$. The dual of the statement is $(q \rightarrow r)^d = \neg q \wedge r$

b) $p \rightarrow (q \vee r) \Leftrightarrow \neg p \vee (q \vee r)$. The dual is $\neg p \wedge (q \wedge r)$

c) $p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$

So, the dual of the statement is $(p \leftrightarrow q)^d = (\neg p \wedge q) \vee (\neg q \wedge p)$.

Example 1.2.12

Write the Dual of the logical equivalence $(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \Leftrightarrow (p \wedge q)$ where p, q and r are primitive statements.

Solution:

The dual of the given statement is $(\neg p \wedge q) \vee (p \vee (p \vee q)) \Leftrightarrow (p \vee q)$

Example 1.3.1

Rita is baking a cake. If Rita is baking a cake, then she is not practicing her flute. If Rita is not practicing her flute, then her father will not buy her a car. Therefore, Rita's father will not buy her a car. Check the validity of the statement.

Solution:

The given propositions are

p : Rita is baking a cake

q : Rita is practicing her flute

r : Rita's father will buy her a car

The argument is

p

$p \rightarrow \neg q$

$\neg q \rightarrow \neg r$

.....
 $\therefore \neg r$

Steps		Reason
1	p	Premises
2	$p \rightarrow \neg q$	Premises
3	$\neg q \rightarrow \neg r$	Premises
4	$p \rightarrow \neg r$	Step (2), (3) Law of Syllogism
5	$\neg r$	Step (1), (4) Modus Ponens

The conclusion follows from the premises. $\therefore$ the argument is valid.

Example 1.3.2

If horses or cows eat grass, then the mosquito is the national bird. If the mosquito is the national bird then peanut butter tastes good on hot dogs. But peanut butter tastes terrible on hot dogs. Therefore, the cow didn't eat grass. Establish the validity of the argument.

Solution:

The given propositions are,

p : Horses eat grass

q : Cows eat grass

r : Mosquito is the national bird

s : Peanut butter tastes good on hot dogs

The hypothesis is

$$p \vee q \rightarrow r$$

$$r \rightarrow s$$

$$\neg s$$

conclusion is: $\therefore \neg q$

Steps		Reason
1	$p \vee q \rightarrow r$	Premises
2	$r \rightarrow s$	Premises
3	$p \vee q \rightarrow s$	Step (1), (2) Law of Syllogism
4	$\neg s$	Premises
5	$\neg(p \vee q)$	Step (3), (4) Modus Tollens
6	$\neg p \wedge \neg q$	Step 5, De Morgan's Law
7	$\neg q$	Step 6, Rule of conjunctive Simplification

The conclusion follows from the premises. $\therefore$ the argument is valid.

Example 1.3.3

If I join IIT then I will get best education. If I get best education, then I will get job in USA. If I get job in USA then I will become a millionaire. Therefore, if I joined in IIT then I will become a millionaire. Show that the conclusion follows from the premises.

Solution:

The given propositions are

p : I join IIT

q : I will get best education

r : I will get job in USA

s : I become a millionaire

The argument is

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ r \rightarrow s \end{array}$$

.....

$\therefore p \rightarrow s$

Steps	Reason
1 $p \rightarrow q$	Premises
2 $q \rightarrow r$	Premises
3 $p \rightarrow r$	Step (1),(2) Law of Syllogism
4 $r \rightarrow s$	Premises
5 $p \rightarrow s$	Step (4) Law of Syllogism

$\therefore$ the conclusion follows from the premises.

Example 1.3.4

If the band could not play rock music or the refreshments were not delivered on time, then the new year party would have been cancelled and Alice would have been angry. If the party were cancelled, then refunds had to be made. No refunds were made. Therefore, the band could play rock music. Establish the validity of the statement.

Solution:

The given propositions are,

p : The band play rock music

q : The refreshments were delivered on time

r : The new year party was cancelled

s : Alice was angry

t : Refunds were made

The given argument is,

$$(\neg p \vee \neg q) \rightarrow (r \wedge s)$$
$$r \rightarrow t$$
$$\neg t$$

.....

$\therefore p$

Steps	Reason
1 $r \rightarrow t$	Premises
2 $\neg t$	Premises
3 $\neg r$	Step (1),(2) Modus Tollens
4 $\neg r \vee \neg s$	Step (3) Rule of disjunctive Amplification
5 $\neg(r \wedge s)$	Step (4) De morgan's Law
6 $(\neg p \vee \neg q) \rightarrow (r \wedge s)$	Premises
7 $\neg(\neg p \vee \neg q)$	Step (5),(6), Modus Tollens
8 $p \wedge q$	Step (7), Demorgan's Law and law of double negation
9 p	Step (8), Rule of conjunctive Simplification

The argument is valid since the conclusion follows from the premises.

Example 1.3.5

Let p, q, r be the statements given as

p : Arjun studies. q : He plays cricket. r : He passes Data Structures.

Let p_1, p_2, p_3 denote the premises.

p_1 : If Arjun studies, then he will pass Data Structures.

p_2 : If he doesn't play cricket, then he will study.

p_3 : He failed Data Structures.

Determine whether the argument $(p_1 \wedge p_2 \wedge p_3) \rightarrow q$ is valid.

Solution:

The argument is,

$$p_1: p \rightarrow r$$

$$p_2: \neg q \rightarrow p$$

$$p_3: \neg r$$

$$\therefore q$$

Steps	Reason
1 $p \rightarrow r$	Premises
2 $\neg q \rightarrow p$	Premises

3	$\neg r$	Premises
4	$\neg q \rightarrow r$	Step (1) (2) Law of Syllogism
5	$\neg(\neg q)$	Step (3), (4) Modus Tollens
6	q	Step (5) Law of double negation

The argument is valid since the conclusion follows from the premises.

Example 1.3.6

If Dominic goes to the racetrack, then Helen will be mad. If Ralph plays cards all night, then Carmela will be mad. If either Helen or Carmela gets mad, then Veronica will be notified. Veronica has not heard from either of these clients, that is Veronica is not notified. Consequently, Dominic did not make it to the racetrack and Ralph didn't play cards all night. Establish the validity of the statement.

Solution:

The given propositions are

- p : Dominic goes to the racetrack
- q : Helen is mad
- r : Ralph plays cards
- s : Carmela is mad
- t : Veronica will be notified

Then the hypotheses are

- $p \rightarrow q$
- $r \rightarrow s$
- $q \vee s \rightarrow t$
- $\neg t$

.....

$$\therefore \neg p \wedge \neg r$$

Steps		Reason
1	$q \vee s \rightarrow t$	Premises
2	$\neg t$	Premises
3	$\neg(q \vee s)$	Step (1),(2) Modus Tollens
4	$\neg q \wedge \neg s$	Step 3 , De Morgan's Law

5	$\neg q$	Step (4) Rule of conjunctive Simplification
6	$\neg s$	Step (4) Rule of conjunctive Simplification
7	$p \rightarrow q$	Premises
8	$\neg p$	Step (5),(7) Modus Tollens
9	$r \rightarrow s$	Premises
10	$\neg r$	Step (6),(9) Modus Tollens
11	$\neg p \wedge \neg r$	Step (8), (10) Rule of Conjunction

The argument is valid.

Example 1.3.7

If Raichel gets the supervisor's position and works hard, then she will get a raise. If she gets the raise, then she will buy a new car. She has not purchased a new car. Therefore, either Raichel did not get the supervisor's position or she did not work hard. Show that the conclusion follows from the premises.

Solution:

The given propositions are

- p : Raichel gets the supervisor's position
- q : Raichel works hard
- r : Raichel gets the raise
- s : Raichel buy a new car

The argument is

$$(p \wedge q) \rightarrow r$$

$$r \rightarrow s$$

$$\neg s$$

.....

$$\therefore \neg p \vee \neg q$$

Steps		Reason
1	$(p \wedge q) \rightarrow r$	Premises
2	$r \rightarrow s$	Premises
3	$\neg s$	Premises
4	$\neg r$	Step (3), (4) Modus Tollens

$$5 \quad \neg(p \wedge q)$$

Step(1), (4) Modus Tollens

$$6 \quad \neg p \vee \neg q$$

Step (5) and (3) De Morgan's Law

The argument is valid.

Example 1.3.8

Establish the validity of the argument

$$p \rightarrow q$$

$$q \rightarrow (r \wedge s)$$

$$\neg r \vee (\neg t \vee u)$$

$$p \wedge t$$

$$\therefore u$$

Solution:

Steps	Reason
1 $p \rightarrow q$	Premises
2 $q \rightarrow (r \wedge s)$	Premises
3 $p \rightarrow (r \wedge s)$	Step (1),(2) Law of Syllogism
4 $p \wedge t$	Premises
5 p	Step (4) Rule of Conjunctive Simplification
6 $r \wedge s$	Step (5) and (3) Modus Ponens
7 r	Step (6) and rule of Conjunctive Simplification
8 $\neg r \vee (\neg t \vee u)$	Premises
9 $(\neg r \vee \neg t) \vee u$	Step (8), Associative Law
10 $\neg(r \wedge t) \vee u$	Step (9) and De Morgan's law
11 t	Step (4) Rule of Conjunctive Simplification
12 $r \wedge t$	Step (7) and (11) and the rule of Conjunction
13 u	Steps (10),(12) and Rule of disjunctive Syllogism

Example 1.3.9

Establish the validity of the argument

$$p \wedge q$$

$$p \rightarrow (r \wedge q)$$

$$r \rightarrow (s \vee t)$$

$$\neg s$$

$$\therefore t$$

Solution:

	Steps	Reason
1	$p \wedge q$	Premises
2	p	Step (1) and Rule of conjunctive Simplification
3	$p \rightarrow (r \wedge q)$	Premises
4	$r \wedge q$	Step (2),(3) Modus Ponens
5	r	Step (4) Rule of conjunctive Simplification
6	$r \rightarrow (s \vee t)$	Premises
7	$s \vee t$	Step (5), (6) and Modus Ponens
8	$\neg s$	Premises
9	t	Step (7),(8) Rule of disjunctive Syllogism

Example 1.3.10

Establish the validity of the argument by the method of proof by contradiction

$$\begin{array}{l}
 \neg p \leftrightarrow q \\
 q \rightarrow r \\
 \neg r \\
 \hline
 \therefore p
 \end{array}$$

Solution:

	Steps	Reason
1	$\neg p \leftrightarrow q$	Premises
2	$(\neg p \rightarrow q) \wedge (q \rightarrow \neg p)$	Step(1)and $(p \leftrightarrow q) \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
3	$\neg p \rightarrow q$	Step (2) Rule of conjunctive Simplification
4	$q \rightarrow r$	Premises
5	$\neg p \rightarrow r$	Step (3),(4) Law of the Syllogism
6	$\neg p$	Premises (assume negation of conclusion)
7	r	Step (5), (6) and Modus Ponens
8	$\neg r$	Premises
9	$r \wedge \neg r$	Step (7), (8) Rule of Conjunction
10	F_0	Step (9) Inverse law
11	$\therefore p$	Step (6),(10) Method of proof by Contradiction

Example 1.3.11

Establish the validity of the argument by the method of proof by contradiction

$$\begin{array}{l}
 r \rightarrow \neg q \\
 r \vee s \\
 s \rightarrow \neg q \\
 p \rightarrow q \\
 \dots\dots\dots \\
 \therefore \neg p
 \end{array}$$

Solution:

Steps	Reason
1 $r \rightarrow \neg q$	Premises
2 $s \rightarrow \neg q$	Premises
3 $(r \vee s) \rightarrow \neg q$	Step (1),(2) Rule for proof by cases
4 $r \vee s$	Premises
5 $\neg q$	Step (3), (4) Modus Ponens
6 $\neg(\neg p)$	Premises (assume negation of conclusion)
7 p	Step (6) Law of Double negation
8 $p \wedge \neg p$	Step (5), (7) Rule of Conjunction
9 F_0	Step (8) Inverse law
10 $\therefore \neg p$	Step (6),(9) Method of proof by Contradiction

Example 1.3.12

Establish the validity of the argument by the method of proof by contradiction

$$\begin{array}{l}
 p \rightarrow (q \wedge r) \\
 \neg(q \wedge s) \\
 r \rightarrow s \\
 \dots\dots\dots \\
 \therefore \neg p
 \end{array}$$

Solution:

Steps	Reason
1 $p \rightarrow (q \wedge r)$	Premises
2 $r \rightarrow s$	Premises
3 $\neg(q \wedge s)$	Premises
4 $\neg(\neg p)$	Premises (assume negation of conclusion)
5 p	Step (4) Law of Double negation
6 $q \wedge r$	Step (1), (5) Modus Ponens

7	q	Step (6) Rule of conjunctive Simplification
8	r	Step (6) Rule of conjunctive Simplification
9	s	Step (2),(8) Modus Ponens
10	$q \wedge s$	Step (7),(9) Rule of Conjunction
11	$\neg(q \wedge s) \wedge (q \wedge s)$	Step (3),(10) Rule of Conjunction
12	F_0	Step (11) Inverse law
13	$\therefore \neg p$	Step (4), (12) Method of proof by Contradiction

Example 1.3.13

Establish the validity of the argument

$$\begin{array}{l}
 u \rightarrow r \\
 (r \wedge s) \rightarrow (p \vee t) \\
 q \rightarrow (u \wedge s) \\
 \neg t \\
 \dots\dots\dots \\
 \therefore q \rightarrow p
 \end{array}$$

Solution:

	Steps	Reason
1	$u \rightarrow r$	Premises
2	$(r \wedge s) \rightarrow (p \vee t)$	Premises
3	$q \rightarrow (u \wedge s)$	Premises
4	$\neg t$	Premises
5	q	Premises (assume)
6	$u \wedge s$	Step (3), (5) Modus Ponens
7	u	Step (6) Rule of conjunctive Simplification
8	s	Step (6) Rule of conjunctive Simplification
9	r	Step (1),(7) Modus Ponens
10	$r \wedge s$	Step (8),(9) Rule of Conjunction
11	$p \vee t$	Step (2),(10) Modus Ponens
12	p	Step (11) (4) Rule of disjunctive Syllogism
13	$\therefore q \rightarrow p$	Step (5), (12)

Example 1.3.14

Establish the validity of the argument

$$p \rightarrow (q \rightarrow r)$$

$$p \vee \neg s$$

$$q$$

$$\therefore s \rightarrow r$$

Solution:

Steps	Reason
1 $p \rightarrow (q \rightarrow r)$	Premises
2 $p \vee \neg s$	Premises
3 q	Premises
4 s	Premises (assume)
5 $\neg s \vee p$	Step (2) commutative property
6 $s \rightarrow p$	Step (5) $p \rightarrow q \Leftrightarrow \neg p \vee q$
7 $s \rightarrow (q \rightarrow r)$	Step (1), (6) Law of the Syllogism
8 $\neg s \vee (q \rightarrow r)$	Step (7) $p \rightarrow q \Leftrightarrow \neg p \vee q$
9 $\neg s \vee (\neg q \vee r)$	Step (8) $p \rightarrow q \Leftrightarrow \neg p \vee q$
10 $(\neg s \vee \neg q) \vee r$	Step (9) Associative Property
11 $\neg(s \wedge q) \vee r$	Step (10) Demorgan's Law
12 $(s \wedge q) \rightarrow r$	Step (11) $p \rightarrow q \Leftrightarrow \neg p \vee q$
13 $s \wedge q$	Step (3) (4) Rule of Conjunction
14 r	Step (12), (13) Modus Ponens
15 $\therefore s \rightarrow r$	Step (4), (14)

Worked Example 1.4

Example 1.4.1

Write the negation of the following statement.

a) If I drive then I will not walk.

Let the primitive statements be p : I drive

q : I will walk.

Then the given statement is $p \rightarrow \neg q$. The negation of this statement is $\neg(p \rightarrow \neg q) \Leftrightarrow p \wedge q$. So the negation is, I will drive and I will walk.

b) He swims if and only if the water is warm.

Let the primitive statements be p : He swims

q : The water is warm

Then the given statement can be written as $p \leftrightarrow q$. The negation of this statement is $\neg(p \leftrightarrow q) \Leftrightarrow p \leftrightarrow \neg q$. So the negation of the given statement is, He swims if and only if the water is not warm.

c) If it is raining then the game is cancelled

Let the primitive statements be p : It is raining

q : game is cancelled.

Then the given statement is $p \rightarrow q$. The negation of this statement is $\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$. Thus the negation of the given statement is it is raining and the game is not cancelled.

d) Paris is in France and London is in England.

Let the primitive statements be p : Paris is in France

q : London is in England.

The given statement can be written as $p \wedge q$. The negation of this statement is $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$. That is Paris is not in France or London is not in England.

e) 9 is greater than 8 and 6 is less than 10.

Let the primitive statements be p : $9 > 8$

q : $6 < 10$

Then the given statement is $p \wedge q$. The negation of this statement is $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$ ie, $9 \leq 8$ or $6 \geq 10$.

f) If John goes to England, then Mary pays for john's shopping money.

Let the primitive statements be p : John goes to England

q : Mary pays for john's shopping money.

Then the given statement is $p \rightarrow q$. The negation of this statement is $p \wedge \neg q$. So, the negation is John goes to England and Mary does not pay for john's shopping money.

Example 1.4.2

If p, q and r are primitive statements, write the converse, inverse and contrapositive of

- a) $p \rightarrow (q \wedge r)$ b) $(p \vee q) \rightarrow r$

Solution:

- a) The converse of $p \rightarrow (q \wedge r)$ is $(q \wedge r) \rightarrow p$
 The inverse of $p \rightarrow (q \wedge r)$ is $\neg p \rightarrow \neg(q \wedge r)$
 The contrapositive of $p \rightarrow (q \wedge r)$ is $\neg(q \wedge r) \rightarrow \neg p$.

- b) The converse of $(p \vee q) \rightarrow r$ is $r \rightarrow (p \vee q)$
 The inverse of $(p \vee q) \rightarrow r$ is $\neg(p \vee q) \rightarrow \neg r$
 The contrapositive of $(p \vee q) \rightarrow r$ is $\neg r \rightarrow \neg(p \vee q)$

Example 1.4.3

Write the converse, inverse and contrapositive of the following statements. Also find the truth value.

- a) If $0 + 0 = 0$ then $1 + 1 = 1$ b) If $9 > 6$ then $4 < 2$

- c) If $-1 < 3$ and $3 + 7 = 10$ then $\sin\left(\frac{3\pi}{2}\right) = -1$

Solution:

- a) Let us consider $p: 0 + 0 = 0$, $q: 1 + 1 = 1$. Since p is true and q is false $p \rightarrow q$ is false.

The converse of $p \rightarrow q$ is $q \rightarrow p$. ie, If $1 + 1 = 1$ then $0 + 0 = 0$ is a true statement.

The inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$, ie, If $0 + 0 \neq 0$ then $1 + 1 \neq 1$ is a true statement.

The contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$ ie, If $1 + 1 \neq 1$ then $0 + 0 \neq 0$ is a false statement.

The truth value is

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
1	0	0	1	0	1	1	0

Table:1.33

- b) If $9 > 6$ then $4 < 2$.

Let us consider $p: 9 > 6$, $q: 4 < 2$. Since p is true and q is false $p \rightarrow q$ is false.

The converse of $p \rightarrow q$ is $q \rightarrow p$ ie, If $4 < 2$ then $9 > 6$ is true.

The inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$, ie, If $9 \leq 6$ then $4 \geq 2$ is true.

The contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$ ie, If $4 \geq 2$ then $9 \leq 6$ is a false statement.

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg p \rightarrow \neg q$	$\neg q \rightarrow \neg p$
1	0	0	1	0	1	1	0

Table:1.34

- c) If $-1 < 3$ and $3 + 7 = 10$ then $\sin\left(\frac{3\pi}{2}\right) = -1$

Let us consider $p: -1 < 3$, $q: 3 + 7 = 10$ and $r: \sin\left(\frac{3\pi}{2}\right) = -1$. Then the given statement is $p \wedge q \rightarrow r$.

The converse of $p \wedge q \rightarrow r$ is $r \rightarrow p \wedge q$. ie, If $\sin\left(\frac{3\pi}{2}\right) = -1$ then $-1 < 3$ and $3 + 7 = 10$.

The inverse is $\neg(p \wedge q) \rightarrow \neg r$, ie, If $-1 \geq 3$ or $3 + 7 \neq 10$ then $\sin\left(\frac{3\pi}{2}\right) \neq -1$.

The contrapositive is $\neg r \rightarrow \neg(p \wedge q)$ ie, If $\sin\left(\frac{3\pi}{2}\right) \neq -1$ then $-1 \geq 3$ or $3 + 7 \neq 10$.

The truth value is

p	q	r	$p \wedge q$	$p \wedge q \rightarrow r$	$r \rightarrow p \wedge q$	$\neg(p \wedge q) \rightarrow \neg r$	$\neg r \rightarrow \neg(p \wedge q)$
1	1	1	1	1	1	1	1

Table:1.35

Example 1.4.4

Negate and simplify the statement

- a) $(p \vee q) \rightarrow r$ b) $p \rightarrow (\neg q \wedge r)$ c) $p \rightarrow (q \rightarrow r)$

Solution:

a) $\neg[(p \vee q) \rightarrow r]$

$$\Leftrightarrow \neg[\neg(p \vee q) \vee r] \text{ since } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow [\neg\neg(p \vee q) \wedge \neg r] \text{ by Demorgan's Law}$$

$$\Leftrightarrow (p \vee q) \wedge \neg r \text{ By law of double negation.}$$

Therefore, the negation of the statement is $(p \vee q) \wedge \neg r$.

b) $\neg(p \rightarrow (\neg q \wedge r))$

$$\Leftrightarrow \neg(\neg p \vee (\neg q \wedge r)) \text{ since } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow \neg\neg p \wedge \neg(\neg q \wedge r) \text{ by Demorgan's Law}$$

$$\Leftrightarrow p \wedge (q \vee \neg r) \text{ By law of double negation and De morgan's Law}$$

Therefore, the negation of $p \rightarrow (\neg q \wedge r)$ is $p \wedge (q \vee \neg r)$

c) $\neg(p \rightarrow (q \rightarrow r))$

$$\Leftrightarrow \neg(p \rightarrow (\neg q \vee r)) \text{ since } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow \neg(\neg p \vee (\neg q \vee r)) \text{ since } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow \neg\neg p \wedge \neg(\neg q \vee r) \text{ by Demorgan's Law}$$

$\Leftrightarrow p \wedge (q \wedge \neg r)$ By law of double negation and De Morgan's Law

Therefore, the negation is $p \wedge (q \wedge \neg r)$.

Example 1.4.5

Express the negation of the statement $p \leftrightarrow q$ in terms of the connectives $\vee$ and $\wedge$.

Solution:

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) \Leftrightarrow (\neg p \vee q) \wedge (\neg q \vee p)$$

$$\text{So, } \neg(p \leftrightarrow q) \Leftrightarrow \neg[(\neg p \vee q) \wedge (\neg q \vee p)] \Leftrightarrow (p \wedge \neg q) \vee (q \wedge \neg p).$$

Exercise: 1.4

1. Write the negation of the following statement.

- If the milk is white then the cow is not black
- If Harold passes his C++ course and finishes his data structure project, then he will graduate at the end of the semester.
- Robin is doing her homework and Kiran is practicing her piano lessons.
- If he studies, he will pass the examination.

2. Write the converse, inverse and contrapositive of the following statement.

- If P is a square, then P is a rectangle.
- If Today is Friday, then tomorrow is Saturday.

3. Negate and simplify the statement

- $(p \wedge q) \rightarrow r$
- $p \vee q \vee (\neg p \wedge \neg q \wedge r)$

4. Write the Dual of a) $p \rightarrow (q \wedge r)$ b) $(p \rightarrow q) \rightarrow r$ c) $p \vee q \vee (q \rightarrow p)$

Answers:

- The milk is white and the cow is black
 - Harold did pass his C++ course and he did finish his data structure project but he did not graduate at the end of the semester.
 - Robin is not doing her homework, or Kiran is not practicing her piano lessons.
 - He studies and he will not pass the examination.
- If P is a rectangle, then P is a square – converse
If P is not a square, then P is not a rectangle- inverse
If P is not a rectangle, then P is not a square- contrapositive
 - If tomorrow is Saturday, then Today is Friday- converse
If today is not Friday, then tomorrow is not Saturday – inverse
If tomorrow is not Saturday, then today is not Friday- contrapositive
- $(p \wedge q) \wedge \neg r$
 - $\neg p \wedge \neg q \wedge \neg r$
- $\neg p \wedge (q \vee r)$
 - $p \wedge \neg q \wedge r$
 - $p \wedge q \wedge (\neg q \wedge p)$

Example 1.5.1

Let $p(x)$: x is a rational number, $q(x)$: x is a real number, and the universe is the set of all real numbers. Check the truth value of the statement $\forall x [(p(x) \rightarrow q(x))]$.

Solution:

Let a be a rational number. Since all rational numbers are real $p(a) \rightarrow q(a)$ is true.

So $\forall x [(p(x) \rightarrow q(x))]$ is true.

Example 1.5.2

Let $p(x)$: $x \leq 3$, $q(x)$: $x + 1$ is odd, and $r(x)$: $x > 0$, the universe be the set of all integers.

Check the validity of the following statements

a) $[p(3) \vee (q(3) \vee \neg r(3))]$ b) $[p(2) \wedge q(2)] \rightarrow r(2)$

Solution:

a) $p(3)$ is true, $q(3)$ is false and $r(3)$ is true. Therefore $p(3) \vee [q(3) \vee \neg r(3)]$ is true.

b) $p(2)$ is true, $q(2)$ is true and $r(2)$ is true. Therefore $[p(2) \wedge q(2)] \rightarrow r(2)$ is true.

Example 1.5.3

Let $p(x)$, $q(x)$ and $r(x)$ denote the following open statements

$p(x): x^2 - 8x + 15 = 0$, $q(x): x$ is odd, $r(x): x > 0$

For the universe of all integers, determine the truth or falsity of each of the following statements. If the statement is false, give a counter example.

a) $\forall x [p(x) \rightarrow q(x)]$ b) $\forall x [q(x) \rightarrow r(x)]$ c) $\exists x [r(x) \rightarrow p(x)]$

d) $\forall x [p(x) \vee q(x) \rightarrow r(x)]$

Solution:

a) The integer 3 and 5 are solutions of the equation $x^2 - 8x + 15 = 0$. If we replace x by 3 or 5 then $p(3)$ and $p(5)$ are always true. $q(3)$ and $q(5)$ are also true. So $p(x) \rightarrow q(x)$ is true.

If we consider an odd integer b other than 3 or 5 then $p(b)$ is false while $q(b)$ is true. $p(b) \rightarrow q(b)$ is true regardless of the truth value of $q(b)$.

If we consider an even integer c then $p(c)$ and $q(c)$ are false. Then also $p(c) \rightarrow q(c)$ is true. Consequently $p(x) \rightarrow q(x)$ is a true statement for all replacement of x taken from the universe of integers. $\therefore \forall x [p(x) \rightarrow q(x)]$ is true.

b) The statement $\forall x [q(x) \rightarrow r(x)]$ is a false statement. By a counter example we can show that the statement is false. Replace x by the integer -1 we can see that $r(-1)$ is false and $q(-1)$ is a true statement. Therefore $q(-1) \rightarrow r(-1)$ is false and consequently $\forall x [q(x) \rightarrow r(x)]$ is false.

c) If we replace x by 3 then $r(3)$ and $p(3)$ are true. Then $r(3) \rightarrow p(3)$ is true. Therefore $\exists x [r(x) \rightarrow p(x)]$ is a true statement.

d) Replace x by the integer -1 we can see that $p(-1)$ is false, $q(-1)$ is true and $r(-1)$ is a false statement. Therefore $[p(-1) \vee q(-1)] \rightarrow r(-1)$ is false and consequently $\forall x [p(x) \vee q(x) \rightarrow r(x)]$ is a false statement.

Example 1.5.4

Let $p(x)$, $q(x)$ and $r(x)$ denote the following open statements

$$p(x): x \geq 0, \quad q(x): x^2 \geq 0, \quad r(x): x^2 - 3x - 4 = 0 \text{ and } s(x): x^2 - 3 > 0$$

For the universe of all real numbers, determine the truth or falsity of each of the following statements. If the statement is false, give a counter example.

- a) $\exists x [p(x) \wedge r(x)]$ b) $\forall x [p(x) \rightarrow q(x)]$ c) $\forall x [q(x) \rightarrow s(x)]$ d) $\forall x [r(x) \vee s(x)]$
 e) $\forall x [r(x) \rightarrow s(x)]$

Solution:

a) The integer 4 is a member of the universe and $p(4)$ and $r(4)$ are true. Therefore,

$\exists x [p(x) \wedge r(x)]$ is a true statement.

b) If we replace x by a positive real number a then $p(a)$ and $q(a)$ are true. Therefore $p(a) \rightarrow q(a)$ is true.

If we replace x by a negative real number b then $p(b)$ is false and $q(b)$ is true. But $p(b) \rightarrow q(b)$ is true regardless of the truth value of $q(b)$. Consequently $p(x) \rightarrow q(x)$ is true for all replacement of x taken from the universe of real numbers. $\forall x [p(x) \rightarrow q(x)]$ is true.

c) If we replace x by a positive real number 1 then $q(1)$ is true and $s(1)$ false. Therefore $q(1) \rightarrow s(1)$ is false. Therefore $\forall x [q(x) \rightarrow s(x)]$ is false.

d) If we replace x by a positive integer 1 then $r(1)$ is false and $s(1)$ false. Therefore $r(1) \vee s(1)$ is false. Therefore $\forall x [r(x) \vee s(x)]$ is a false statement.

e) If we replace x by a negative integer -1 then $r(-1)$ is true and $s(-1)$ false. Therefore $r(-1) \rightarrow s(-1)$ is false. Therefore $\forall x [r(x) \rightarrow s(x)]$ is a false statement.

Example 1.5.5

Find the inverse, converse and contrapositive of the statement 'If $m > n$ then $m^2 > n^2$ ' where the universe is the set of all positive integers.

Solution:

Let $p: m > n$, $q: m^2 > n^2$. Then the given statement is $\forall x [p(x) \rightarrow q(x)]$ is a true statement.

The converse of the given statement is $\forall x [q(x) \rightarrow p(x)]$ is a true statement since if $m^2 > n^2$ then $m > n$ is true for the set of all positive integers.

The inverse of the given statement is $\forall x [\neg p(x) \rightarrow \neg q(x)]$ is a true statement since if $m \leq n$ then $m^2 \leq n^2$ is true for the set of all positive integers.

The contrapositive of the given statement is $\forall x[\neg q(x) \rightarrow \neg p(x)]$ is a true statement since if $m^2 \leq n^2$ then $m \leq n$.

Example 1.5.6

Find the inverse, converse and contrapositive of the statement 'All squares are equilateral' where the universe is the set of all quadrilaterals in the plane.

Solution:

Let $s(x)$: x is a square, $e(x)$ = x is equilateral. The given statement is $\forall x[s(x) \rightarrow e(x)]$ is a true statement.

The converse of the given statement is $\forall x[e(x) \rightarrow s(x)]$ is false since all equilaterals are not squares.

The inverse of the given statement is $\forall x[\neg s(x) \rightarrow \neg e(x)]$ is false.

The contrapositive of the given statement is $\forall x[\neg e(x) \rightarrow \neg s(x)]$ is a true statement.

Example 1.5.7

Find the inverse, converse and contrapositive of the statement 'If the magnitude of a real number is greater than 3, then the number itself is greater than 3' where the universe is the set of all real numbers.

Solution:

Let $p(x)$ and $q(x)$ be the open statements $p(x): |x| > 3$, $q(x): x > 3$. The given statement is $\forall x[p(x) \rightarrow q(x)]$ is a false statement since $p(-5)$ is true while $q(-5)$ is false.

So, $[p(-5) \rightarrow q(-5)]$ is a false statement. Therefore $\forall x[p(x) \rightarrow q(x)]$ is a false statement.

The inverse of the given statement is $\forall x[\neg p(x) \rightarrow \neg q(x)]$ is a true statement since the magnitude of a real number is less than or equal to 3 then the number itself is less than or equal to 3.

The converse of the given statement is $\forall x[q(x) \rightarrow p(x)]$ is a true statement since every real number greater than 3 has magnitude greater than 3.

The contrapositive of the given statement is $\forall x[\neg q(x) \rightarrow \neg p(x)]$ is a false statement because if we replace x by -5 then the statement is false.

Example 1.5.8

Find the inverse, converse and contrapositive of the statement $\forall x[(x > 3) \rightarrow (x^2 > 9)]$ where the universe is the set of all real numbers.

Solution:

Let $p(x)$ and $q(x)$ are open statements $p(x): x > 3$, $q(x): x^2 > 9$. The given statement is $\forall x[p(x) \rightarrow q(x)]$ is true.

The inverse of the given statement is $\forall x[\neg p(x) \rightarrow \neg q(x)]$ is a false statement since $p(-5)$ is true while $q(-5)$ is false. So, $[p(-5) \rightarrow q(-5)]$ is false. Therefore $\forall x[p(x) \rightarrow q(x)]$ is a false statement.

The converse of the given statement is $\forall x[q(x) \rightarrow p(x)]$ is a false statement since $q(-5)$ is true and $p(-5)$ is false. So, $[p(-5) \rightarrow q(-5)]$ is false. Therefore $\forall x[q(x) \rightarrow p(x)]$ is a false statement.

The contrapositive of the given statement is $\forall x[(x^2 \leq 9 \rightarrow (x \leq 3))]$ is a true statement.

Exercise:1.5

1. Let $p(x): x^2 = 2x$ where universe is the set of all integers. Determine whether the following statement is true or false

- a) $p(0)$ b) $p(2)$ c) $\exists x p(x)$ d) $\forall x p(x)$

2. Let $p(x)$, $q(x)$ and $r(x)$ denote the following open statements

$$p(x): x^2 - 7x + 10 = 0, \quad q(x): x^2 - 2x - 3 = 0, \quad r(x): x < 0$$

For the universe of all integers, determine the truth or falsity of each of the following statements. If the statement is false, give a counter example.

- a) $\forall x[p(x) \rightarrow \neg r(x)]$ b) $\forall x[q(x) \rightarrow r(x)]$
c) $\exists x[q(x) \rightarrow r(x)]$ d) $\exists x[p(x) \rightarrow r(x)]$

3. Let $p(x)$ and $q(x)$ denote the following open statements

$p(x): x > 1$, $q(x): x < 6$. For the universe of all integers, determine the truth or falsity of each of the following statements

- a) $\forall x[p(x) \vee q(x)]$ b) $\forall x[p(x)] \vee \forall x[q(x)]$ c) $\exists x[p(x) \wedge q(x)]$

4. Find the inverse, converse and contrapositive of the statement $\forall x[p(x) \rightarrow (r(x) \vee q(x))]$

$p(x): |x| > 3$, $q(x): x > 3$, $r(x): x < -3$ which is defined for the universe of all real numbers.

5. Let $p(x, y)$, $q(x, y)$ denote the following open statements,

$p(x, y): x^2 \geq y$, $q(x, y): x + 2 < y$. If the universe consists of all real numbers, determine the truth value of the following statement,

- a) $p(2, 4)$ b) $p(-3, 8) \vee q(1, 3)$ 3) $p(2, 2) \rightarrow q(1, 1)$

Answers:

1. a) True b) True c) True d) False
2. a) True b) False when $x = 3$, $r(3)$ is false c) True d) True
3. a) True b) False c) True

4. The inverse of the given statement is $\forall x[\neg p(x) \rightarrow \neg(r(x) \vee q(x))]$ is false.

The converse of the given statement is $\forall x[(r(x) \vee q(x)) \rightarrow p(x)]$ is true.

The contrapositive of the given statement is $\forall x[\neg(r(x) \vee q(x)) \rightarrow \neg p(x)]$ is a false statement.

5. a) True b) False c) False

Worked Example 1.6

Example 1.6.1

Let $p(x): 2x + 1 = 5$, $q(x): x^2 = 9$ Prove that $\exists x [p(x) \wedge q(x)]$ is not logically equivalent to $[\exists x p(x) \wedge \exists x q(x)]$.

Solution:

If we replace x by an integer a then $p(a): 2a + 1 = 5$, and $q(a): a^2 = 9$ then $a = 2$ and $a = \pm 3$ which shows that $p(a) \wedge q(a)$ is a false statement. $\therefore \exists x [p(x) \wedge q(x)]$ is false because there is no integer a such that $2x + 1 = 5$ and $x^2 = 9$. But there is an integer $b = 2$ such that $2b + 1 = 5$ and there is an integer $c = 3$ or -3 such that $x^2 = 9$. $\therefore [\exists x p(x) \wedge \exists x q(x)]$ is true. ie, $\exists x [p(x) \wedge q(x)] \not\equiv [\exists x p(x) \wedge \exists x q(x)]$.

Example 1.6.2

Consider $p(x, y): x + y = 17$. Show that $\forall x \exists y p(x, y) \not\equiv \exists y \forall x p(x, y)$ where the universe is the set of all integers.

Solution:

If we replace x by any integer a then $y = 17 - a$ does exist and $a + 17 - a = 17$. Therefore $\forall x \exists y p(x, y)$ is a true statement.

Consider $\exists y \forall x p(x, y)$ the statement is false because if we select a value of y as $y = a$ then the only value of x is $17 - a$ which satisfy the statement $x + y = 17$.

Therefore $\forall x \exists y p(x, y) \not\equiv \exists y \forall x p(x, y)$.

Example 1.6.3

Determine the truth value of each statement if the universe comprises all nonzero integers.

- a) $\exists x \exists y [xy = 1]$ b) $\exists x \forall y [xy = 1]$ c) $\forall x \exists y [xy = 1]$

d) $\exists x \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$ d) $\exists x \exists y [(3x - y = 7) \wedge (2x + 4y = 3)]$

Solution:

a) For some $x = 1$ and some $y = 1$, we have $xy = 1 \times 1 = 1$ is true. $\therefore \exists x \exists y [xy = 1]$ is true.

b) For some $x = 2$ and for $y = 3$, $xy = 2 \times 3 \neq 1$. $\therefore \exists x \forall y [xy = 1]$ is false.

c) For some $y = 2$ and $x = 1$, $xy = 2 \times 1 \neq 1$. $\therefore \forall x \exists y [xy = 1]$ is false.

d) Solving $2x + y = 5$ and $x - 3y = -8$, we get $x = 1$ and $y = 3$. $\therefore$ the statement is true for $x = 1$ and $y = 3$. $\therefore \exists x \exists y [(2x + y = 5) \wedge (x - 3y = -8)]$ is true.

e) Solving $3x - y = 7$ and $2x + 4y = 3$, we get $x = 31/14$ and $y = -5/14$ which are not integers.

$\therefore \exists x \exists y [(3x - y = 7) \wedge (2x + 4y = 3)]$ is a false statement.

Example 1.6.4

Determine the truth value of each statement if the universe comprises all real numbers.

- a) $\forall x [x^2 > x]$ b) $\exists x [If\ x > 1\ then\ x^2 > x]$ c) $\forall x \forall y [x^2 < y + 1]$
 d) $\exists x \exists y [x^2 < y + 1]$ e) $\forall x \forall y [x^2 + y^2 = 9]$ f) $\exists x \exists y [x^2 + y^2 = 9]$
 g) $\exists x \forall y [x^2 + y^2 \geq 0]$

Solution:

a) If we replace $x = \frac{1}{3}$, $x^2 < x$, the statement is false.

b) The statement is true since when $x = 0$ the statement is true.

c) When $x = 3$ and $y = 2$, $\forall x \forall y [x^2 < y + 1]$ is false.

d) When $x = 1$ and $y = 1$, $\exists x \exists y [x^2 < y + 1]$ is true

e) When $x = 2$ and $y = 3$, $\forall x \forall y [x^2 + y^2 = 9]$ is false.

f) When $x = \sqrt{8}$ and $y = 1$, $\exists x \exists y [x^2 + y^2 = 9]$ is true

g) When $x = 0$ and for all values of y , $\exists x \forall y [x^2 + y^2 \geq 0]$ is true.

Example 1.6.5

Negate the statement 'If x is odd then $x^2 - 1$ is even' where the universe is the set of all integers

Solution:

Let $p(x)$: x is odd, $q(x)$: $x^2 - 1$ is even. Then the given statement is $\forall x[p(x) \rightarrow q(x)]$ is true.

The negation of this statement is determine as follows,

$$\neg[\forall x(p(x) \rightarrow q(x))] \Leftrightarrow \exists x[\neg(p(x) \rightarrow q(x))]$$

$$\Leftrightarrow \exists x[\neg(\neg p(x) \vee q(x))] \text{ since } p \rightarrow q \Leftrightarrow \neg p \vee q$$

$$\Leftrightarrow \exists x[(\neg\neg p(x) \wedge \neg q(x))] \text{ Demorgan's Law}$$

$$\Leftrightarrow \exists x[(p(x) \wedge \neg q(x))] \text{ Law of double negation}$$

Thus the negation of the given statement is 'there exist an integer x such that x is odd and $x^2 - 1$ is odd.

Example 1.6.6

Negate the statement $\exists x[(p(x) \wedge q(x))]$ where $p(x)$: $2x + 1 = 5$, $q(x)$: $x^2 = 9$ and the universe is the set of all integers.

Solution:

The negation is

$$\neg\exists x[(p(x) \wedge q(x))] \Leftrightarrow \forall x[\neg(p(x) \wedge q(x))]$$

$$\Leftrightarrow \forall x[\neg p(x) \vee \neg q(x)] \text{ Demorgan's Law}$$

That is for every integer x , $2x + 1 \neq 5$ or $x^2 \neq 9$ is the negation of the given statement.

Exercise: 1.6

1. Determine the truth value of each statement if the universe comprises all nonzero integers

- a) $\exists x \exists y [xy = 2]$ b) $\exists x \forall y [xy = 2]$ c) $\forall x \exists y [xy = 2]$
 d) $\exists x \exists y [(3x + y = 8) \wedge (2x - y = 7)]$ e) $\exists x \exists y [(4x + 2y = 3) \wedge (x - y = 1)]$

2. Determine the truth value of each statement if the universe comprises all real numbers.

- a) $\forall x [x^2 > x]$ b) $\forall x \forall y [x^2 < y + 1]$ c) $\exists x \exists y [x^2 < y + 1]$
 d) $\forall x \forall y [x^2 + y^2 = 9]$ e) $\exists x \forall y [x^2 + y^2 \geq 0]$

3. Negate the statement 'If x is a real number where $x^2 > 16$ then $x < -4$ or $x > 4$ where the universe is the set of all real numbers.

Answers:

1. a) True b) False c) False d) True e) False

2. a) when $x = \frac{1}{3}$ the statement is False b) when $x = 3$ and $y = 2$ the statement is False

c) True d) when $x = 2$ and $y = 3$ the statement is False e) True

3. For some real x , $x^2 > 16$ but $-4 \leq x$ and $x \leq 4$

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